

SeokHyeon (Lucas) Kong

CONTACT INFORMATION	Moore 1st floor University of Pennsylvania 200 South 33rd Street, Philadelphia, PA 19104	Personal Website: lucasshkong.github.io Email: lucasshkong@gmail.com
RESEARCH INTERESTS	My research interests lie in <i>Quantum Computer Architecture</i> , currently focusing on Quantum Error Correction.	
EDUCATION	University of Pennsylvania , Philadelphia, PA • Ph.D. in Computer and Information Science • Advisor: Prof. Gushu Li	Aug 2026
	Sungkyunkwan University (SKKU) , Suwon, Korea • B.S. in Electronic and Electrical Engineering • B.S. in Advanced Semiconductor Engineering – System Architecture Track • <i>Graduated with Summa Cum Laude</i> (GPA: 4.36 / 4.5)	Mar 2020 – Feb 2026
	The Pennsylvania State University , University Park, PA • Exchange Student in Computer Engineering • <i>Dean's List</i> (Grade: 4.0 / 4.0)	Fall 2024
PUBLICATIONS	[1] Characterizing the System Overhead of Discrete Gaussian Noise Generation for Differential Privacy SeokHyeon Kong, Donghwan Kim, Euiseong Seo, and Kiwan Maeng CAL 2025 – IEEE Computer Architecture Letters	
	[2] Predicting Key Regional Real Estate Prices Using Machine Learning Technique: with an emphasis on Jeonsae system SeokHyeon Kong and Hayoung Oh JKIICE 2023 – Journal of the Korean Institute of Information and Communication Engineering	
POSTER PRESENTATION	[3] Solving Constraint Satisfaction Problems with Variational Quantum Algorithm on NISQ Devices SeokHyeon Kong, Chanhee Park, Jang-il Moon, and Denny Dahl Qcenter – Quantum Information Research Support Center, Korea	
EXPERIENCE	Hanwha Systems , Seongnam, Korea <i>Software Engineer in the Avionics System R&D Team</i> • Collaborated with The Boeing Company on ELAD (Eagle Large Area Display) for F-15 aircraft.	Jan 2026 – Jul 2026
	IonQ , Remote <i>Research Mentee with Dr. Denny Dahl</i> • Addressed the Instant Insanity problem by employing a graph-theoretic encoding and a divide-and-conquer approach with two distinct post-processing methods [3]. • Solved the Four Corners Map Coloring problem using a Quantum Approximate Optimization Algorithm (QAOA) with two different encoding schemes [3].	Aug 2025 – Sep 2025

	The Pennsylvania State University , University Park, PA Aug 2024 – Sep 2025 <i>Undergraduate Researcher with Prof. Kiwan Maeng</i>	
	- Conducted GPU profiling with Nsight Compute and Nsight Systems to analyze performance bottlenecks and system-level overhead in differential privacy [1]. - Proposed an optimization that resolved the dominant Geometric operation overhead, reducing it from 50% to 19% while preserving Gaussian noise quality [1].	
TECHNICAL PROJECT	SoC Design and Practice , SKKU <i>Graduate Level Coursework</i>	Fall 2025
	- Designed and implemented a multi-channel, pipelined DMAC engine and cache controller in SystemVerilog with AMBA APB-based configuration and AXI-based data transfer. - Synthesized and verified the design using Synopsys Design Compiler, optimizing area and timing to meet target performance specifications.	
TEACHING EXPERIENCE	Academic Tutor , SKKU <i>Intro to Electromagnetism</i>	Fall 2023
	Led weekly Q&A sessions and problem-solving discussions.	
	Teaching Assistant , SKKU <i>Engineering Mathematics 2</i>	Summer 2023
	Prepared and organized class materials, including exams.	
HONORS AND AWARDS	Graduate Fellowship and Research Stipend Award , UPenn Summa Cum Laude , SKKU Director's Award – Quantum Hackathon, QCenter President's List , SKKU Dean's List , SKKU Korea–U.S. Student Exchange Scholarship in the field of high-tech industry (\$9,000), KIAT Student Success Scholarship , SKKU	2026 2026 2025 2024, 2025 SP2023, FA2023, SP2024, FA2024, SP2025 FA2024 FA2023
SKILLS	Software: Qiskit, CUDA, C, Python, MATLAB, LaTeX Hardware: System Verilog, Verdi, Virtuoso, Nsight Systems, Nsight Compute	
LEADERSHIP AND SERVICE	Executive Board , President's List (Honor Society at SKKU) Qiskit Advocate , IBM Quantum Squad Leader , Republic of Korea Army (Military Service)	2025 2025 Apr 2021 – Oct 2022
REFERENCES	Gushu Li, Ph.D. Email: gushuli@seas.upenn.edu Assistant Professor of Computer and Information Science University of Pennsylvania Kiwan Maeng, Ph.D. Email: kvm6242@psu.edu Assistant Professor of Computer Science and Engineering The Pennsylvania State University Denny Dahl, Ph.D. Email: denny.dahl@ionq.co Customer Success Director IonQ	